

Case Study Report

Design and Creative Technologies

Torrens University, Australia

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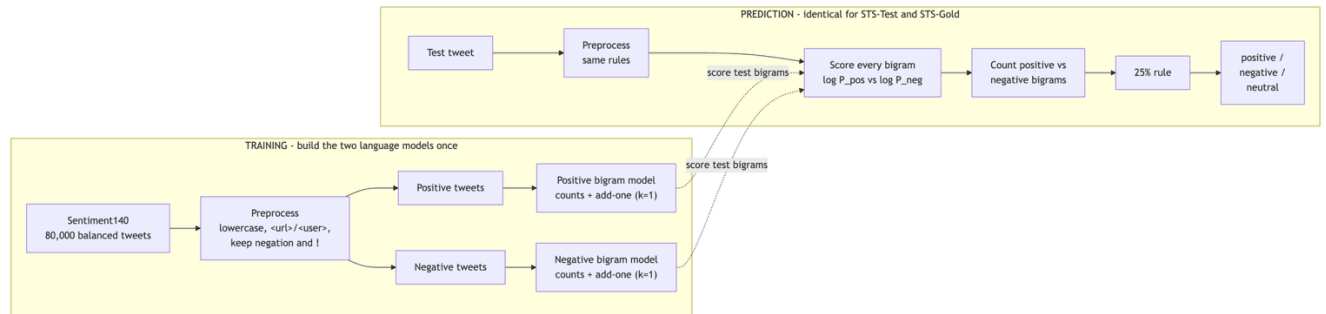
1. Introduction

Twitter sentiment analysis assigns positive, negative, or neutral sentiment to short, noisy social-media messages. Zhao, Gui and Zhang (2018) approached this with deep convolutional neural networks; this assessment instead implements the required probabilistic N-Gram language model. Separate positive and negative bigram models are trained from labelled tweets, and a simple threshold rule decides each tweet's sentiment. The same model, preprocessing, and rule are applied to two of the paper's five datasets – STS-Test and STS-Gold – so their outcomes can be compared on identical terms.

2. Implementation method

The classifier uses a bigram model ($n = 2$) with add-one (Laplace) smoothing over a 49,038-token vocabulary, trained on a balanced 80,000 tweet sample of the Sentiment140 corpus. Tweets are normalized by lowercasing, replacing URLs and usernames with placeholder tokens, stripping the hash symbol while keeping the hashtag word, and retaining exclamation marks and negation. For each test tweet, every bigram is scored under both models; a bigram counts as positive or negative when its smoothed conditional log-probability is higher under that model. Following the brief, a tweet is labelled positive when at least one quarter of its bigrams are positive and outnumber the negative ones, negative by the mirror rule, and neutral otherwise.

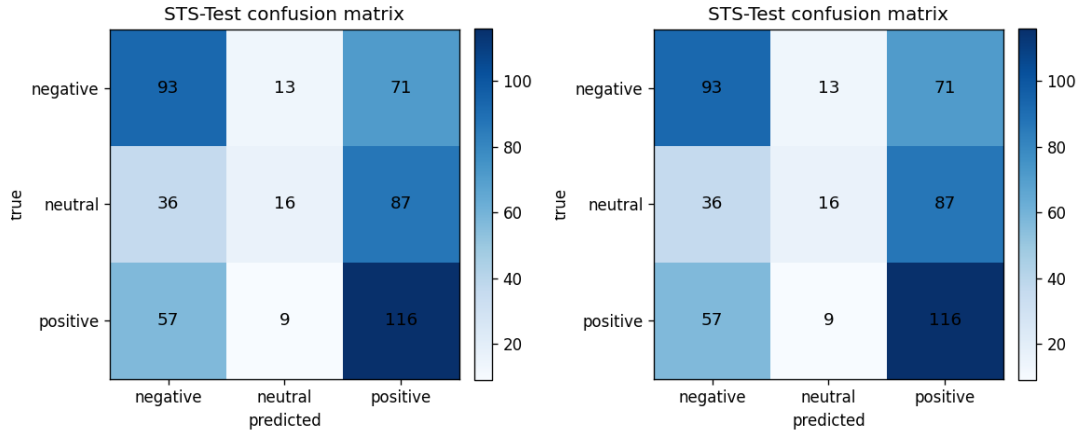
Figure 1 – Bigram model's training and prediction pipeline.



3. Results and comparison

The two sources produced clearly different outcomes (Table 1). On STS-Gold the model reached 0.72 accuracy and 0.73 macro-F1; on STS-Test it managed only 0.45 accuracy and 0.40 macro-F1. Three factors explain the gap. First, STS-Gold is a two-class problem, whereas STS-Test adds a neutral class the model struggles with: it predicted neutral for only 38 of 498 tweets, far below the 139 truly neutral ones. With Laplace smoothing almost every bigram leans slightly positive or negative, so the 25% threshold is easily crossed and neutral is rarely chosen. Second, STS-Gold's informal tweet style closely matches the emotion-labelled training corpus, giving denser, more reliable bigram evidence. Third, both datasets share a positive-leaning bias – the predicted positives exceed true positives in each – inherited from the training data. The shared model behavior is therefore consistent, but its usefulness depends heavily on the target source. A trigram variant performed worse on both datasets (0.42 and 0.55 accuracy), confirming that higher-order N-Grams are too sparse for short tweets, so the bigram is the submitted model.

Figure 2 – Confusion matrices illustrating the model's performance.



4. Critical Reflection

The errors reveal the core limitation of N-Grams. Sarcasm ("worked on 5 bone marrow cases today! All + for cancer!"), emoji sentiment, and long-range negation ("no. it is too big. I'm quite happy") are misread, because local word-pair counts carry no contextual meaning and unseen phrases receive only smoothed probabilities. The model also has no learned concept of neutrality; that class emerges purely from the threshold. These weaknesses are exactly what the deep CNN of Zhao, Gui, and Zhang (2018) addresses through learned word embeddings and wider context.

5. Conclusion

The N-Gram model is a transparent, reproducible baseline that classifies tweets reasonably when training and test language overlap, as on STS-Gold, but degrades on harder, multi-class sources such as STS-Test. Its accuracy depends on dataset vocabulary, class balance, and preprocessing rather than genuine language understanding.

6. Appendices

6.1. Appendix A – Glossary of Terms

Table 4 – Term and Definition support table.

Term	Definition
AUC	Area Under the Receiver Operating Characteristic Curve; a performance metric for classification models measuring the ability to distinguish between classes. Values range from 0.5 (random) to 1.0 (perfect).
DHITA	Digital Health Intervention Text Analytics; the two-step NLP framework proposed by Funk et al. (2020) for automated feature extraction and predictive modeling from DHI text data.
DHI	Digital Health Intervention; technology-based programs (apps, online platforms, web therapy) designed to support health outcomes at scale.
Federated Learning	A distributed machine learning approach where models are trained across multiple decentralised data sources without sharing raw data, preserving participant privacy.
GloVe	Global Vectors for Word Representation; a method for generating static word embeddings by analysing word co-occurrence statistics across a large corpus.
LDA	Latent Dirichlet Allocation; a generative probabilistic model for topic modeling that represents each document as a mixture of latent topics.
LASSO	Least Absolute Shrinkage and Selection Operator; a regression method applying L1 regularisation that drives irrelevant feature coefficients to zero, performing feature selection and regularisation simultaneously.
LIME	Local Interpretable Model-agnostic Explanations; a technique that explains individual predictions of any classifier by approximating it locally with an interpretable model.
Logistic Regression	A statistical model that predicts the probability of a binary outcome using a logistic (sigmoid) function applied to a linear combination of input features.

NLP	Natural Language Processing; a subfield of AI concerned with enabling computers to analyse, understand, and generate human language.
POS Tagging	Part-of-Speech Tagging; the process of labelling each word in a text with its grammatical category (noun, verb, adjective, etc.), used as a feature in NLP pipelines.
Random Forest	An ensemble learning method that constructs many decision trees on random data subsets and outputs the majority vote across trees, reducing overfitting through aggregation.
RCT	Randomised Controlled Trial; an experimental study design where participants are randomly assigned to intervention or control conditions to establish causal evidence.
ROC	Receiver Operating Characteristic curve; a plot of true positive rate against false positive rate at varying classification thresholds, used to evaluate binary classifier performance.
SHAP	SHapley Additive exPlanations; a game-theory-based method for explaining individual model predictions by attributing a contribution value to each input feature.
Transformer	A deep learning architecture using self-attention mechanisms to process sequential data in parallel, enabling rich contextual understanding in NLP tasks (e.g., BERT, BioBERT, MentalBERT).
Sentiment Lexicon	A curated list of words associated with specific emotions or sentiment polarities, used for sentiment analysis.

6.2. Appendix B – Results

Table 1 – Results (bigram model, identical settings)

Metric	STS-Test	STS-Gold
Records evaluated	498	2,034
True negative / neutral / positive	177 / 139 / 182	1,402 / 0 / 632
Predicted negative / neutral / positive	186 / 38 / 274	1,110 / 136 / 788
Accuracy	0.452	0.719
Macro-F1	0.401	0.726

End of Appendix Section

Statement of Acknowledgment

I acknowledge that I have used the following AI tool(s) in the creation of this report:

- OpenAI ChatGPT (Codex-5.5)
- Anthropic Claude (Opus 4.8)

These tools were used to assist with understanding N-Gram probabilistic language models and the Markov assumption, structuring the Python classifier pipeline (preprocessing, bigram counting, add-one smoothing, and the 25% decision rule), debugging code and improving comment quality, interpreting the comparison between the two datasets, improving the clarity of academic language, and supporting APA 7th referencing conventions.

Prompt examples:

1. *"Explain why add-one (Laplace) smoothing is needed in a bigram language model and how it changes the conditional probability formula."*
2. *"My classifier predicts neutral for very few tweets - how does the 25% threshold rule interact with smoothing to produce that, and how should I explain it critically in the report?"*
3. *"Format this as APA 7th: Go, Bhayani & Huang (2009), Twitter sentiment classification using distant supervision, Stanford CS224N project report."*

I confirm that the use of the AI tool has been in accordance with the Torrens University Australia Academic Integrity Policy and TUA, Think and MDS's Position Paper on the Use of AI. I confirm that the final output is authored by me and represents my own critical thinking, analysis, and synthesis of sources. I take full responsibility for the final content of this report.

7. References

- Go, A., Bhayani, R., & Huang, L. (2009). *Twitter sentiment classification using distant supervision* (CS224N Project Report, Stanford).
<https://cs.stanford.edu/people/alecmgo/papers/TwitterDistantSupervision09.pdf>
- Saif, H., Fernández, M., He, Y., & Alani, H. (2013). Evaluation datasets for Twitter sentiment analysis: A survey and a new dataset, the STS-Gold. *Proceedings of the 1st International Workshop on Emotion and Sentiment in Social and Expressive Media (ESSEM)*.
<https://oro.open.ac.uk/40660/>
- Zhao, J., Gui, X., & Zhang, X. (2018). Deep convolution neural networks for Twitter sentiment analysis. *IEEE Access*, 6, 23253–23260. <https://doi.org/10.1109/ACCESS.2017.2776930>